CS307 Fall 26 Project 1

General Requirement:

- It is a group project with **only 2 teammates** who are **in the same lab session**. Each group should finish the project independently and submit only one report written by the teammates.
    - The teammate you select for Project 1 will also be your teammate for Project 2. It is not allowed to change teammates once paired.
- You should submit the report before the deadline. All late submissions after the deadline will receive a score of zero.
- DO NOT copy ANY sentences and figures from the Internet and your classmates. Plagiarism is strictly prohibited in this course.
- The text description should be rigorous, the overall design should be logical organised, the report structure and the layout of diagram should be clear and easy to read, otherwise, you will receive a penalty in the scoring stage.
- The number of pages for your report should be between **5 and 7**. Reports **only or less than 5 pages** and **more than 7 pages** will receive a penalty in the scoring stage.

DBMS can help us manage data in a convenient manner and improve the efficiency of data retrieval. Your work of Project 1 is mainly divided into three parts below:

1. Design an E-R diagram based on the provided data file and data relationships.
2. Design a relational database using PostgreSQL according to the provided data file.
3. Import all data into the database.
4. Perform Create, Read, Update, and Delete (CRUD) operations using the database

Background

This project centers on the **Shenzhen urban rail transit (metro) line information of 2026**. It systematically compiles the stations of all operating and under-construction lines, their administrative districts, opening years and expected opening years, and first/last train times; it also organizes each station's entrances/exits and the nearby key landmarks (hospitals, schools, shopping malls, parks and attractions, government offices, high-speed-rail hubs, control points/ports, the airport, museums, etc.). Two structured datasets are produced — a station master table and a station-exit landmark table — for easy lookup, comparison, and downstream use.

The collection and consolidation drew on public materials and assistance provided by **TREA SOLO, Doubao, and ChatGPT**. All content is based on real operating and planned information; **where any external source disagrees with the two project files, `shenzhen_stations.csv` (station master table) and `station_exits.csv` (station-exit landmark table) take precedence, and external information must not override the established figures in the files.**The Report and your Tasks

Basic Information of Your Group [Report Part]

1. Names, student IDs, and the lab session of the group members
2. You are required to write down the contributions and the percentages of contributions for each group member. **Please clearly state which task(s)/part of the task(s) is/are done by which member in the group.**
    - **If you failed to link a task/part of a task to one of the group members, we will not count the score for the task** (since we don't know who accomplished this task; maybe it was done by an elf while you were sleeping at night?).

About one page

Task 1: E-R Diagram (20%) [Report Part]

Make an E-R Diagram of your database design with any diagram software. Hand-drawn results will not be accepted. Please follow the standard of E-R diagrams.

In the report, you are required to provide a snapshot of the E-R diagram. Also, please specify the name of the software/online service you use for drawing the diagram.

About one page

Task 2: Database Design (20%) [Report Part]

Design the tables and columns based on the background provided above. Generate the E-R diagram via the **Show Visualization** feature if you are using **DataGrip** (if you plan to use other GUI database clients, please find the corresponding way by yourself to generate the E-R diagram according to the client you use.). Briefly describe the design of the tables and columns including (but not limited to) the meanings of tables and columns

In the report, you are required to provide the following content:

1. Attach the snapshot of the E-R diagram generated by DataGrip.
2. Briefly describe the table designs and the meanings of each table and column.

In addition, please submit an **SQL file** as an attachment that contains the **DDL**s (`create table` statements) for all the tables you created. **Please make it into a separate file but not copy and paste the statements into the report.**

About one or two pages

Notes for the database design:

Following requirements you must be accomplished.

1. All data items should base on csv file.
2. Your design needs to follow the requirements of the **three normal forms**.
3. Every row in each table should be uniquely identified by its primary key. (You may use a simple or a composite primary key).
4. Every table should be involved in a foreign key. No isolated table is allowed. (每个表要有外键，或者有其他表的外键指向。)

5. Your design should contain no circular foreign-key links. (对于表之间的外键方向，不能有环。例如：A表有外键关联B表，B表有外健关联C表，C表有外健关联A表)
6. You should use appropriate data types for different fields.

Try to complete the following requirements

1. Each table should contain at least one mandatory (“Not Null”) column (including the primary key but not the id column).
2. Other than the system-generated self-increment ID column, there should be at least one column with the "unique" constraint. (除了主键自增的id之外，需要有其他unique约束的列)
3. Your design should be easy to expand when requirements change.

Task 3: Data Import (30%)

In this task, you should write scripts to import the content in `csv` into the database you have designed before. After importing the data, you should also make sure all data is successfully imported.

Task 3.1 Basic Requirements: 5% [Report Part]

Scripts & AI assistance used to import data (table)

Introduce **all** scripts and AI tools used to import / prepare the data. For every item, fill one row in the table below.

For example:

Name	Type	Author	Description (function + how to run)
`script1.py`	Script	Your name	Run `python script1.py` to read `raw.csv`, clean fields, and import passenger data into the `passenger` table.
`script2.sql`	SQL script	Your name	Executed to create target tables and load the intermediate file.
ChatGPT (model: GPT-4o)	LLM	Your name	Helped reformat the CSV-file to import by datagrip
Doubao (agent)	AI Agent	Your name	Helped generate the table-creation SQL and insertion logic;

About one or two pages

Task 3.2 Generate a One-Click Import SQL Script (5%) [Presetation Part]

Submit a standalone, complete PostgreSQL script (e.g., `import.sql`) that supports one-click import and is run live during the defense.

The script must contain:

- **All `CREATE TABLE` statements**: the complete structure of every table — column types, primary keys, foreign keys / inter-table relationships, `NOT NULL` and `UNIQUE` constraints, as well as any indexes needed for later queries.
- **All `INSERT` statements**: every data record, with values mapped correctly to the corresponding columns.
- **Conflict-free and repeatable execution (idempotent)**: re-running the script on the same database must not produce duplicate data or errors. Begin with a deterministic cleanup (`DROP TABLE IF EXISTS ...`) so that re-execution always returns to a clean state.
- Make the data import as fast as possible.

提交一个**独立、完整的 PostgreSQL 脚本**（如 `import.sql`），**答辩现场运行，能实现一键导入。**

脚本需包含：

1. **全部建表语句（CREATE TABLE）**：每张表的完整结构 —— 字段类型、主键、外键 / 表间关系、`NOT NULL`、`UNIQUE` 约束，以及后续查询所需的索引。
2. **全部插入语句（INSERT）**：每条数据记录，字段与列正确对应。
3. **避免冲突、可重复执行（幂等）**：同一库上重复运行不产生重复数据不报错。 开头做确定性清理（`DROP TABLE IF EXISTS ...`），保证重跑回到干净状态。
4. **想办法让导入数据尽可能快** 。

Task 3.3 Data Accuracy checking: 20% [Presetation Part]

According to the data in csv file, we will **give several questions** in presentation week to check whether all data have been correctly imported into your database. Please prepare SQL queries for the following requirements **in the report**, and you will run the queries by **datagrip** in this part **during your presentation week**.

15% for accuracy

5% for Whether to prepare SQL statements before and complete the process of accuracy checking on time

After data imported, in presentation time, we will test about following questions.

Q1. How many metro stations are there in total?

> 439

Q2. How many metro lines are there in total?

> 29

Q3. Output the number of stations with status "Operation" for each district, sorted by district name.

	district	cnt
1	Bao'an	58
2	Dapeng	1
3	Futian	52
4	Guangming	24
5	Longgang	70
6	Longhua	21
7	Luohu	32
8	Nanshan	67
9	Pingshan	16
10	Yantian	9

Q4. Output the number of stations opened in each different year on a **given English name of metro line**, sorted by opening year in ascending order.

For `Line 5`

	opening	cnt
1	2011	27
2	2019	7
3	2025	3

Q5. **Given the English name of a metro line**, output the English station name, and the corresponding number of exits for the station(s) with the most exits on that line. If multiple stations tie for first place in exit count, return all of them.

For `Line 1`

	english_name	cnt
1	Shenzhen University	10
2	Chegongmiao	10

Q6. **Given a prefix match of a landmark** (like 'xxx%'), return the station's Chinese name, its district, the corresponding exit, and the full landmark name. Sort the results by English name.

For `塘朗%`

	chinese_name	district	exit	name
1	深大丽湖	Nanshan	C	塘朗大厦
2	深大丽湖	Nanshan	C	塘朗新村
3	塘朗	Nanshan	D	塘朗村
4	塘朗	Nanshan	B	塘朗地铁车辆段
5	塘朗	Nanshan	A	塘朗城
6	桃源村	Nanshan	D	塘朗山公园

Q7. **Given an English name of station, an English name of metro line and a number n**, query the name of the station n stops forward and the station n stops backward along that line.

For Q7 question, create a **Function** named `get_station_n_neighbor` and to passing three paramaters `station_name`, `line_name` and `n` , then return the result as below:

For example.

```
select get_station_n_neighbor('Mawan', 'Line 5', 5);
```

	get_station_n_neighbor
1	*<null>*
2	Linhai

```
select get_station_n_neighbor('Mawan', 'Line 5', 2);
```

	get_station_n_neighbor
1	Liwan
2	Qianwan

这里使用现场导入后的数据测试。

Q1. 总共多少个地铁站?

Q2. 总共多少个地铁线?

Q3. 输出每个区的状态为"Operation" 的地铁站数量, 根据district 名字排序。

Q4. **给定一个地铁线英文名**， 输出不同年份开通的地铁站的数量，根据opening升序排序。

Q5. **给定一个地铁线英文名**，输出某个地铁线出口最多的地铁站英文名，以及对应出口数量，如果多个地铁站出口数量并列第一多，那么这些数据都要返回。

Q6. 给定一个地标**前通配符**模糊查找(like 'xxx%')， 返回地铁站的中文名、区、对应出口，以及地标全称，结果按照地铁站英文名排序。

Q7. 给出一个地铁站的英文名、与一个地铁线的英文名，与一个数字n，查询地铁站在一个地铁线路上向前数第n站与向后数第n站的地铁站名字。

对于Q7这道题目，需要创建一个名字叫做 `get_station_n_neighbor` 的**Function**, 并传递3个参数：`station_name`, `line_name` and `n` ， 返回结果如截图所示。

Task 4: Perform Create, Read, Update, and Delete (CRUD) operations using the database. (30%) [Presetation Part]

In this section, 5% of the grade is allocated to your presentation skills, which includes your familiarity with the system, your communication skills, and your time management abilities.

Basic Requirements (15%)

1. Add, modify, delete and search a metro station.
2. Add, modify, delete and search a metro line.
3. Station–line relationship:
    - Place one or more stations into a specified position on a metro line.
    - Remove a station from a metro line.

1. 添加、修改、删除、查询一个地铁站
2. 添加、修改、删除、查询一个地铁线
3. 地铁站与地铁线的关系：

- 将一个或多个地铁站在放入一个地铁线中指定的位置上。
- 将一个地铁站从一个地铁线中移除

Advanced Requirements (10%)

1. Support path query between two stations
2. Given two landmarks, query the metro line(s) that connect them.
3. Query the fare between two stations.
4. Design a visually appealing user-facing interface.
5. Comprehensive code design, e.g., all functions can be completed with a single program launch, and the program has a reasonably complete architectural hierarchy.

1. 支持两个地铁站之间的路径查询
2. 输入两个地标，查询地铁线路
3. 查询两个地铁站之间的票价
4. 整体可以实现出一个面向用户的端口页面形式
5. 综合代码设计：所有的功能可以在一个程序中完整的运行，同时程序的架构层级设计的合理。

Overall requirements in this section:

All functions in this part require the design of command-line prompt output statements.

For example:

```
Please select an operation:
1. Add a metro station.
2. Modify a metro station.
3. Delete a metro station.
4. Search a metro station.
4. Add / modify / delete / Search a metro line.
5. Place a station on a line.
6. Remove a station from a line.
7. Query the n-th station forward / backward.

>> Input:
   1
Please enter the station name:

>> Input:
   梨园
Please enter the English name, district, and status:

>> Input:
   Liyuan, Longgang, Operation

Station "梨园 (Liyuan)" added successfully.

...
```

How to Submit Your Report

Submit the report in PDF format with necessary attachments (such as SQL scripts and source code files) on the bb website before **23:55 on Novermber 15th, 2026, Beijing Time (UTC+8)**. For attachments, please put them into separate directories based on the task, and compress them into a `.zip` archive.